

## SAFETY PRECAUTIONS

|                                     |                              |
|-------------------------------------|------------------------------|
| <input checked="" type="checkbox"/> | Stopped engine               |
| <input checked="" type="checkbox"/> | Block the starting mechanism |
| <input checked="" type="checkbox"/> | Shut off starting air supply |
| <input checked="" type="checkbox"/> | Engage turning gear          |
| <input checked="" type="checkbox"/> | Shut off cooling water       |
| <input checked="" type="checkbox"/> | Shut off fuel oil            |
| <input checked="" type="checkbox"/> | Shut off lubricating oil     |
| <input type="checkbox"/>            | Lock turbocharger rotors     |

## Data

| Ref.                   | Description                                                                                                                                                                                                                                                                                                            | Value                  | Unit |             |          |             |                                                            |             |                  |        |            |  |  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------|-------------|----------|-------------|------------------------------------------------------------|-------------|------------------|--------|------------|--|--|
| D03-01                 | Lifting tool screws, tightening torque                                                                                                                                                                                                                                                                                 | 150                    | Nm   |             |          |             |                                                            |             |                  |        |            |  |  |
| D03-03                 | Cylinder diameter, new                                                                                                                                                                                                                                                                                                 | 500                    | mm   |             |          |             |                                                            |             |                  |        |            |  |  |
| D03-04                 | Piston cleaning ring diameter, new                                                                                                                                                                                                                                                                                     | 498.3                  | mm   |             |          |             |                                                            |             |                  |        |            |  |  |
| D03-05                 | Cylinder liner, complete                                                                                                                                                                                                                                                                                               | 1820                   | kg   |             |          |             |                                                            |             |                  |        |            |  |  |
| D03-06                 | Cooling jacket                                                                                                                                                                                                                                                                                                         | 170                    | kg   |             |          |             |                                                            |             |                  |        |            |  |  |
| D03-16                 | Piston cleaning ring radial width, new                                                                                                                                                                                                                                                                                 | 38.8                   | mm   |             |          |             |                                                            |             |                  |        |            |  |  |
| D03-17                 | PC-ring/Liner max. wear deviation                                                                                                                                                                                                                                                                                      | 0.5                    | mm   |             |          |             |                                                            |             |                  |        |            |  |  |
| D03-18                 | PC-ring replacement criteria:                                                                                                                                                                                                                                                                                          |                        |      |             |          |             |                                                            |             |                  |        |            |  |  |
|                        | <table><tr><td colspan="2">Liner diameter: 500 mm</td></tr><tr><td>Liner wear:</td><td>Install:</td></tr><tr><td>0 – 0.50 mm</td><td>Standard PC-ring<br/>(new or max. wear of diameter 0.50 mm)</td></tr><tr><td>0.50 – 1 mm</td><td>Oversize PC-ring</td></tr><tr><td>&gt; 1 mm</td><td>Dummy ring</td></tr></table> | Liner diameter: 500 mm |      | Liner wear: | Install: | 0 – 0.50 mm | Standard PC-ring<br>(new or max. wear of diameter 0.50 mm) | 0.50 – 1 mm | Oversize PC-ring | > 1 mm | Dummy ring |  |  |
| Liner diameter: 500 mm |                                                                                                                                                                                                                                                                                                                        |                        |      |             |          |             |                                                            |             |                  |        |            |  |  |
| Liner wear:            | Install:                                                                                                                                                                                                                                                                                                               |                        |      |             |          |             |                                                            |             |                  |        |            |  |  |
| 0 – 0.50 mm            | Standard PC-ring<br>(new or max. wear of diameter 0.50 mm)                                                                                                                                                                                                                                                             |                        |      |             |          |             |                                                            |             |                  |        |            |  |  |
| 0.50 – 1 mm            | Oversize PC-ring                                                                                                                                                                                                                                                                                                       |                        |      |             |          |             |                                                            |             |                  |        |            |  |  |
| > 1 mm                 | Dummy ring                                                                                                                                                                                                                                                                                                             |                        |      |             |          |             |                                                            |             |                  |        |            |  |  |

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

| Plate  | Item No. | Description                       |
|--------|----------|-----------------------------------|
| P90251 | 51       | Lifting tool for cylinder liner   |
| P90361 |          | Measuring tool for cylinder liner |
| P90362 |          | Cylinder Liner - Extra Tool       |
| P90366 |          | Cylinder Liner - Lifting Tools    |

1. Dismount the cylinder cover, piston and PC-ring.

*See Procedure 901-1.2.*

*See Procedure 902-1.2.*

Clean the cylinder liner and scavenge air ports.

2. Position the cylinder liner measuring tool.

Measure the cylinder liner with an inside micrometer at the positions indicated on the measuring tool. *See Data.*

3. Take measurements in the fore-and-aft and athwartship directions.

4. Check and assess the condition of the cylinder liner according to the description given in Volume I, Chapter 707.

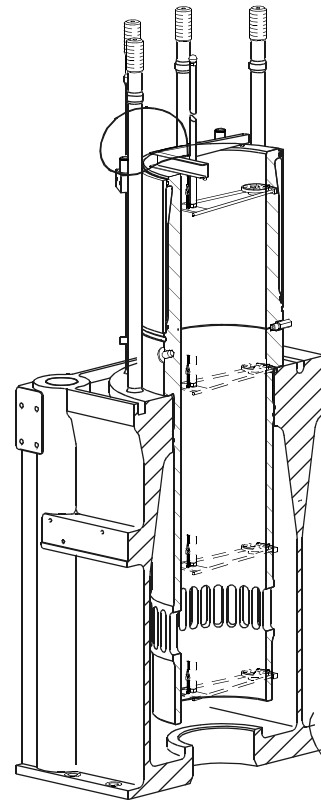
5. Carefully scratch over any scores or marks on the cylinder liner running surface, and grind away the wear ridges. *See Procedure 903-1.3.*

6. Check the lubricating points of the cylinder by pumping cylinder oil to each individual lubricating point.

Clean any blocked lubricating duct.

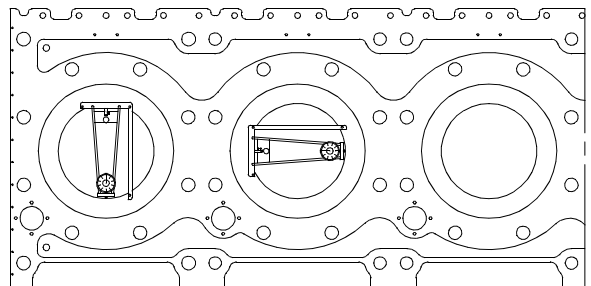
1.

D03-03



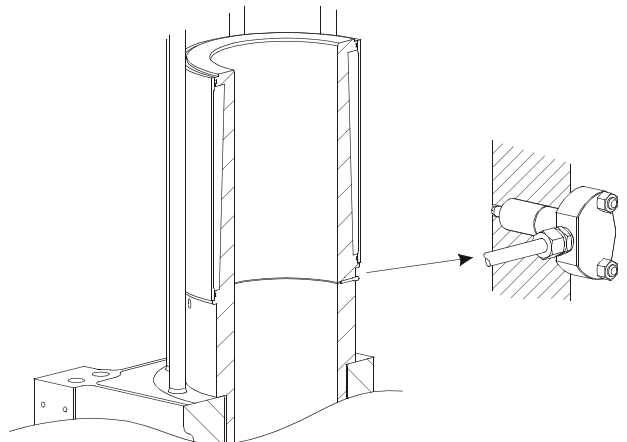
DM903-2.0 90 01

2.

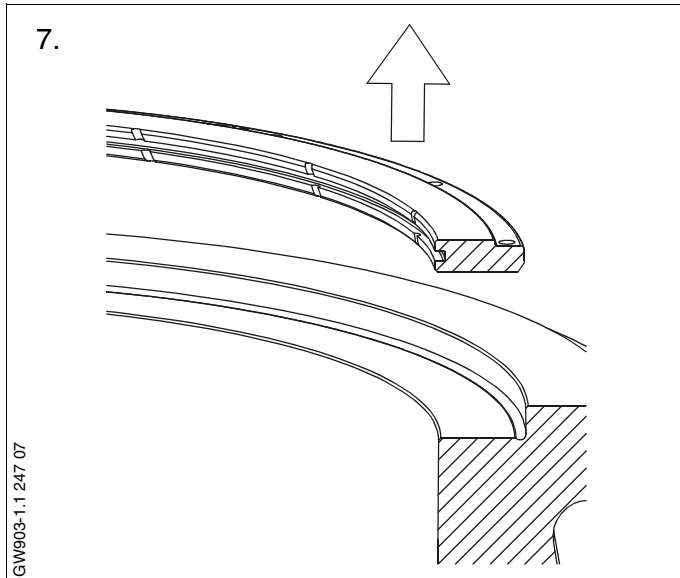


GN903-1.1 203 03

4.



GM903-1.1 253 04

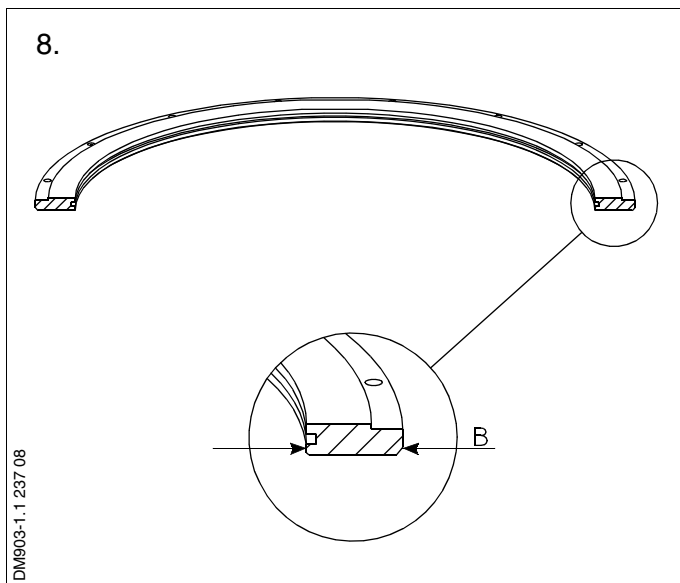


#### 7. Piston cleaning (PC) ring:

The PC ring is to be regarded as an integrated part of the liner and it is intended to follow the service life of the liner.

During inspection of the piston and liner, the PC ring must also be inspected.

When the piston is removed, the PC-ring must always be taken up before the piston. See Procedure 902-1.2.



#### 8. Measure the PC ring radial width B, and find the most worn place. Compare with data D03-16 and calculate the wear of the PC ring. Multiply the wear by 2 to get the PC ring diameter wear.

Measure the liner in the fore-and-aft and athwartship directions at piston skirt TDC position, corresponding to measuring point No. 5 on the liner measuring tool. Compare the measurements with the diameter of a new liner as stated in data. Calculate the wear of the liner.

Compare the PC ring diameter wear with the wear of the liner. If wear of the two components does not deviate more than D03-17, then reinstall the PC ring.

If the deviation between PC ring wear and liner wear is more than D03-17, or if the PC ring is broken or cracked, replace the ring with a new PC ring, an oversize ring or a dummy ring according to the table in D03-18 on the Data sheet.

If it is needed to install an oversize PC ring, contact MAN B&W Diesel for advice.

When a new liner is installed, also a new PC ring must be installed. When a new liner is ordered, also a new PC ring must be ordered.

1. Dismount the cylinder cover.  
*See Procedure 901-1.2.*

Discard the sealing ring from the top of the cylinder liner.

Turn the piston down far enough to make it possible to remove the wear ridges at the top of the liner.  
*See Procedure 903-1.3.*

Dismount the piston and the PC-ring.  
*See Procedure 902-1.2.*

Mount the two lifting screws in the cylinder liner.

### Note!

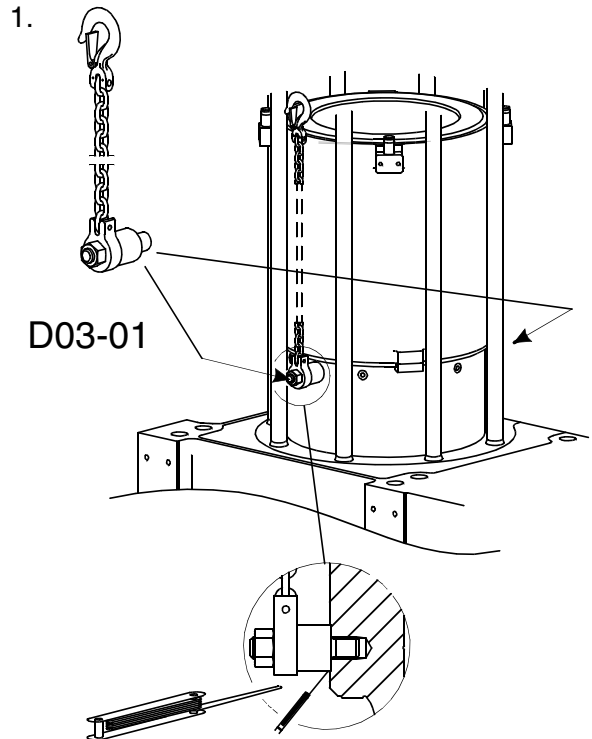
Tighten the lifting screws to the torque stated in Data.

Check with a 0.05 mm feeler gauge that there is no clearance between the screws and the cylinder liner.

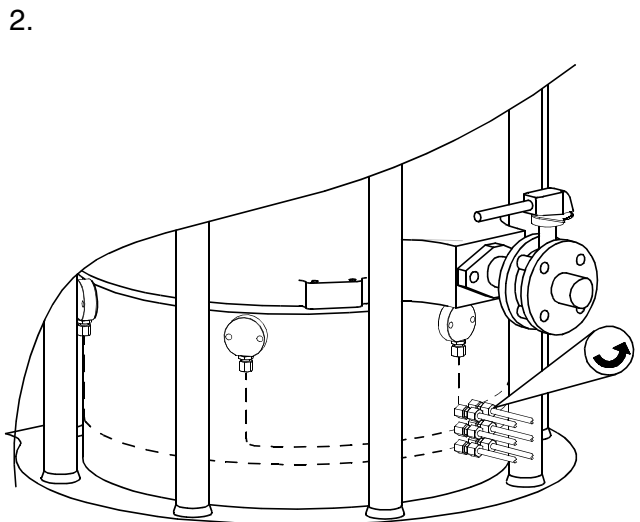
This must be done whenever the lifting tools are mounted on the liner.

2. Disconnect – at the exhaust side of the cylinder – the oil pipes leading from the cylinder lubricator to the non-return valves.

Dismount the four cooling water pipes – between the cooling jacket and cylinder cover – and clean them carefully.

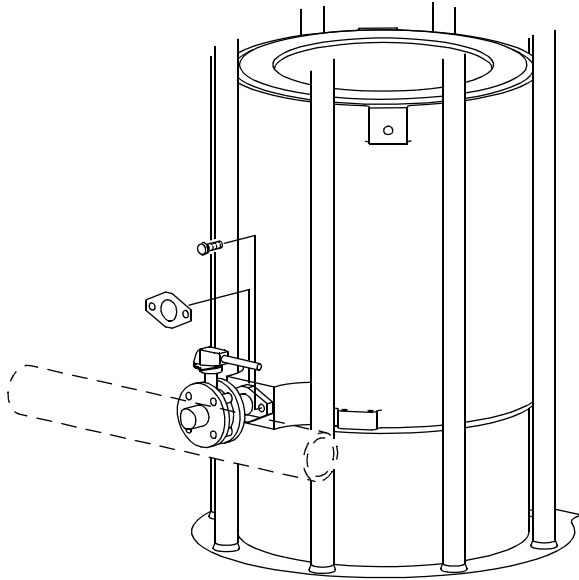


GN903-1.2 224 01



GN903-1.2 224 02

3.



3. Remove the screws of the cooling water inlet pipe.
4. Attach the crane hook to the lifting cross-bar.

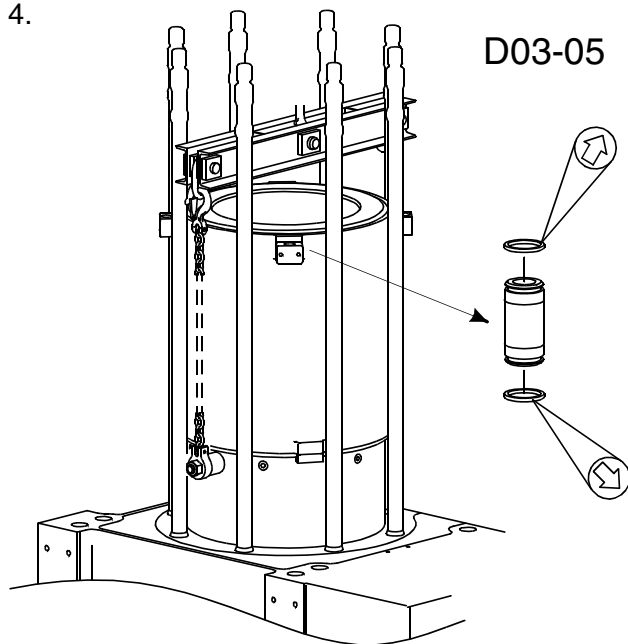
Hook the chains from the lifting tools on to the crossbar and lift the cylinder liner with the cooling jacket out of the cylinder frame.

**Note!**

Low lifting height in the engine room may require the removal of one or more cylinder cover studs before dismantling the cylinder liner.

4.

D03-05



Land the cylinder liner vertically on, for instance, a couple of planks.

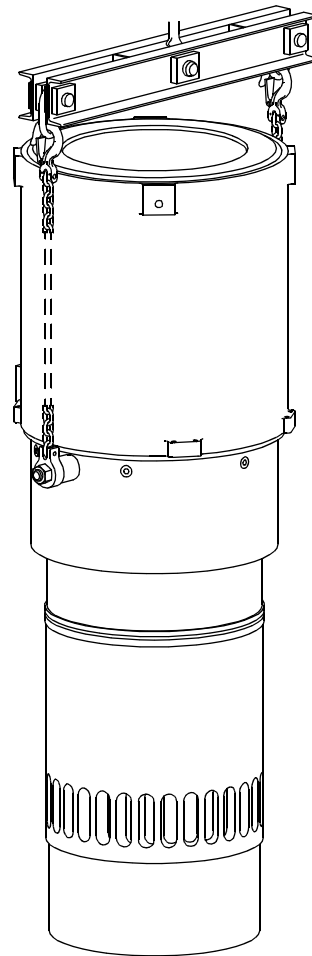
Clean the cylinder frame internally, taking special care with the contact surfaces for the cylinder liner at the top of the cylinder frame.

Discard the O-rings on the cooling water pipes.

Clean the pipes carefully.

5. The lifting screws are also used for transporting the cylinder liner.

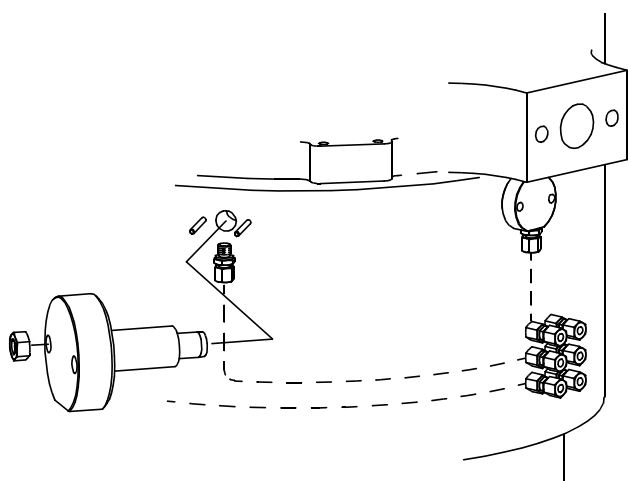
5.



D03-05

HN903-1.2 244 05

1.



1. Unscrew the nuts on the non-return valves and remove the valves.

Clean and inspect the non-return valves.

2. Attach two tackles to the crossbar, as shown.

Mount two lifting eye bolts in the cooling jacket.

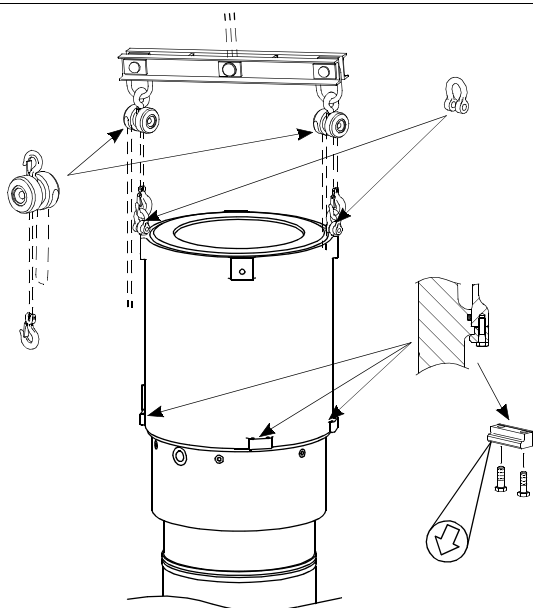
Hook the tackles on to the lifting eye bolts on the cooling jacket and haul tight.

Remove the two lifting screws from the cylinder liner.

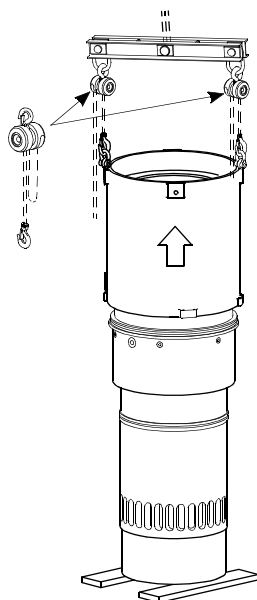
Remove the four clamps which fix the cooling jacket to the cylinder liner.

3. Lift the cooling jacket away by means of the tackles, and land it on the wooden planks.

2.



3.



D03-06

4. Check and assess the condition of the cylinder liner according to the description given in Volume I, Chapter 707.

Carefully scratch over any scores or marks on the cylinder liner running surface, by means of a rough grindstone held in the hand.

Check in the top of the liner for a wear ridge (where the piston rings reverse direction). If there is any sign of a wear ridge, it is necessary to create a groove by grinding. The groove serves to prevent the build-up of a new wear ridge.

5. It is recommended to use a wear ridge milling machine to create the groove. *For use of the milling machine, see supplier's instruction.*
6. Alternatively, place an old piston ring on the top of the piston and turn to a position that enables the grinding disc to rest on the old piston ring while removing the wear ridge. Use a grinding disc with a round edge.

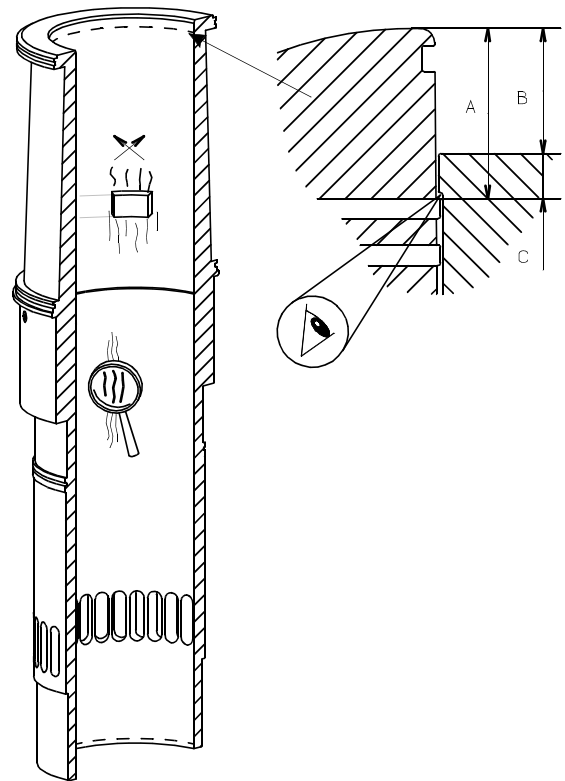
Check the distance **C** before grinding to ascertain that the wear ridge was caused by the top ring at TDC.

The maximum depth of the groove is calculated as  $S_{\max.} = D \times 0.0045$ .

### Note!

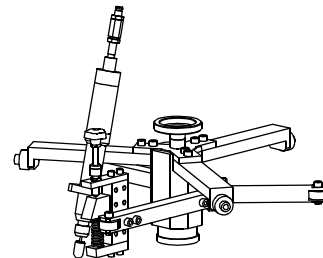
It is of the utmost importance that the groove is made with a regular rounding as shown in the sketch.

4.



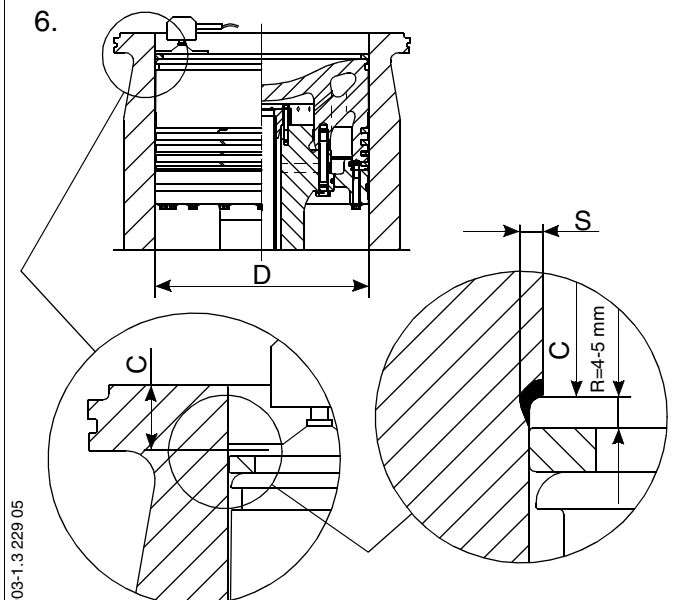
HN903-1.3 221 04

5.



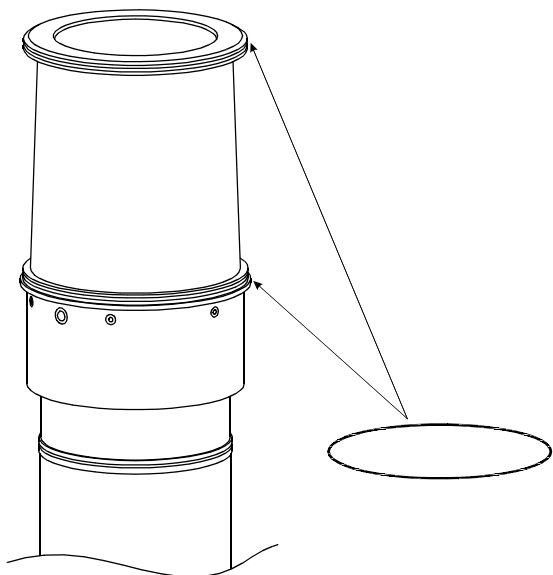
MB903-1.3 231 04

6.



MN903-1.3 229 05

7.



7. Replace the two uppermost O-rings on the cylinder liner.

8. Lower the cooling jacket on to the cylinder liner.

### Note!

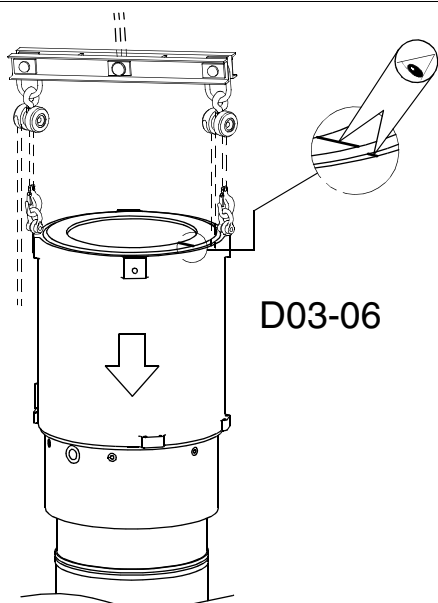
Make sure that the scratch marks in the fuel pump side of the cylinder liner and cooling jacket coincide.

9. Mount the four clamps in the groove of the liner, and tighten the screws.

Remove the tackles from the lifting cross-bar and the cooling jacket lifting eye bolts.

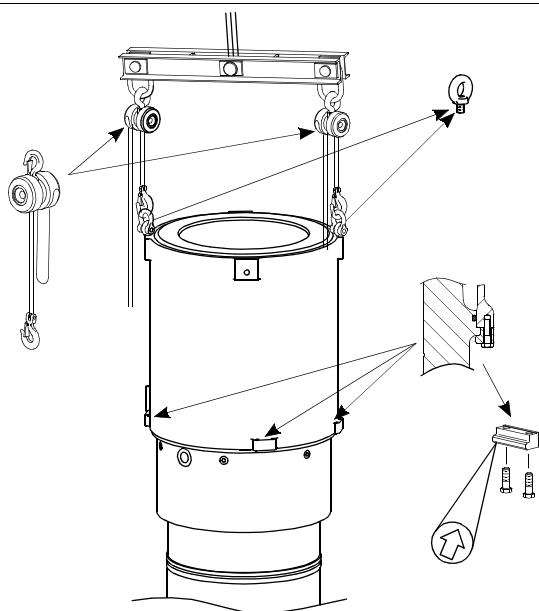
10. Mount the non-return valves for cylinder lubrication. Tighten the nuts and connect the lubricating oil pipes.

8.



D03-06

9.



1. Loosen the water connections on the cooling jacket.

Mount the two lifting screws in the cylinder liner. See *Data D03-01*.

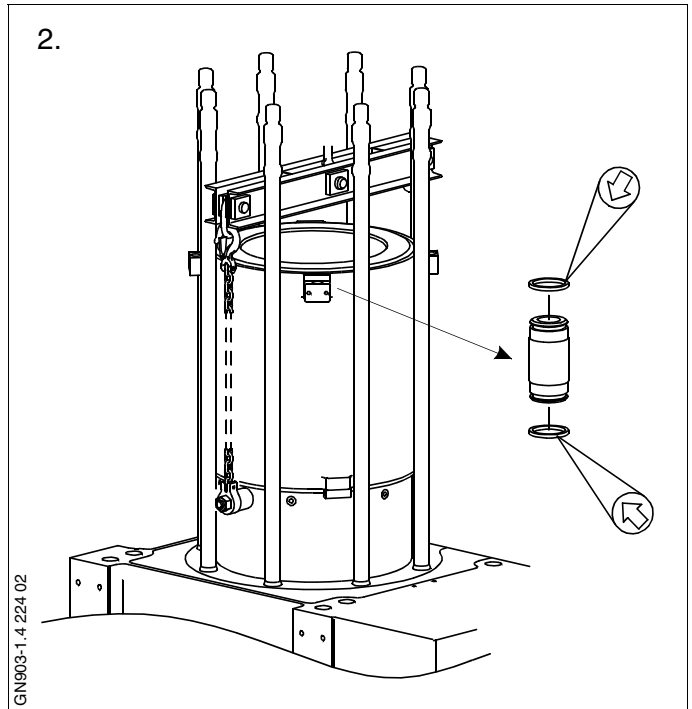
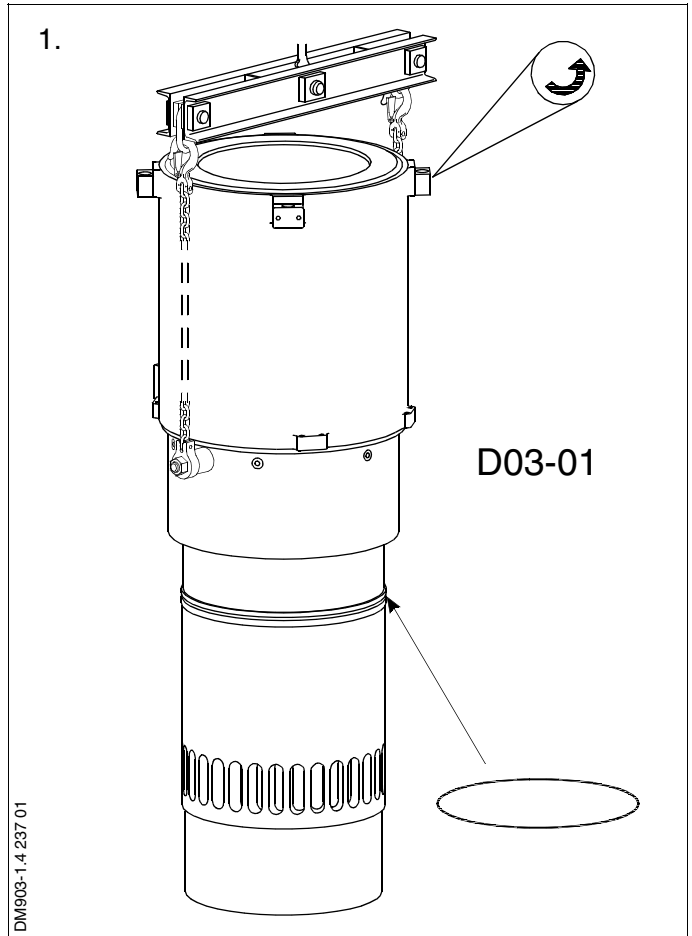
Hook the chains from the lifting crossbar on to the lifting screws, and lift the jacket/liner assembly.

Mount the lowermost O-ring and apply a little lubricating oil on the ring.

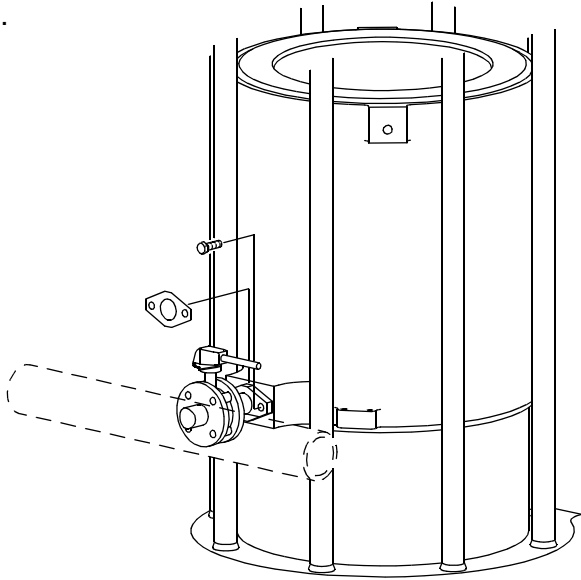
2. Check that the joint surfaces on the cylinder frame and cylinder liner are completely clean.

Coat the joint surfaces with permatex or a similar liquid sealing compound.

Mount the cylinder liner in the cylinder frame. Replace the O-rings on the water connections and mount the water connections on the cooling jacket.



3.



3. Fit a new gasket between the cooling water inlet pipe and the cooling jacket. Mount and tighten the screws.

4. Connect and tighten the oil pipes for the cylinder lubrication.

Vent the cylinder lubricating system until oil, without air bubbles, comes out from the non-return valves. Check that there is no leakage from the system.

When this is in order, check that each individual lubricating point functions correctly.

5. Lubricate the upper part inside of the cylinder liner with cylinder lubricating oil and mount the piston and the PC ring.

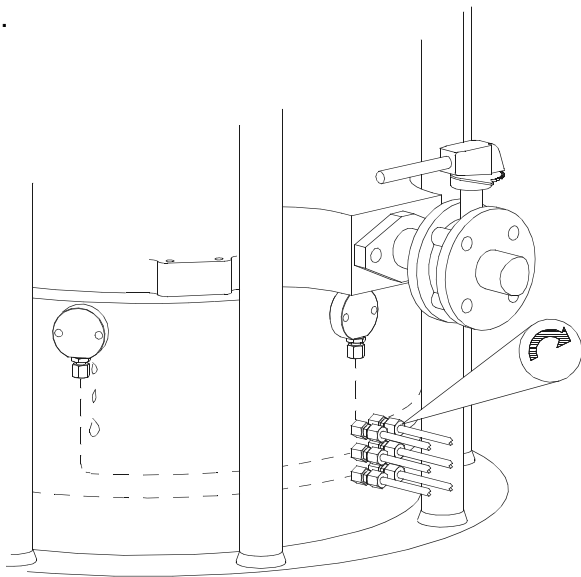
*See Procedure 902-1.4.*

6. Mount the sealing ring and the cylinder cover.

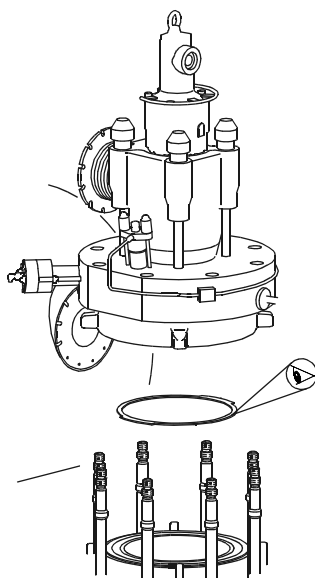
*See Procedure 901-1.4.*

Tighten the upper water connections on the cooling jacket as soon as the cylinder cover is correctly positioned.

4.



6.



## SAFETY PRECAUTIONS

|                                     |                              |
|-------------------------------------|------------------------------|
| <input checked="" type="checkbox"/> | Stopped engine               |
| <input checked="" type="checkbox"/> | Block the starting mechanism |
| <input checked="" type="checkbox"/> | Shut off starting air supply |
| <input checked="" type="checkbox"/> | Engage turning gear          |
| <input type="checkbox"/>            | Shut off cooling water       |
| <input type="checkbox"/>            | Shut off fuel oil            |
| <input type="checkbox"/>            | Shut off lubricating oil     |
| <input type="checkbox"/>            | Lock turbocharger rotors     |

## Data

| Ref.   | Description              | Value | Unit |
|--------|--------------------------|-------|------|
| D03-22 | Minimum crankthrow angle | 103   | °    |
| D03-23 | Maximum crankthrow angle | 114   | °    |
| D03-24 | Optimum crankthrow angle | 108   | °    |



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

| Plate | Item No. | Description |
|-------|----------|-------------|
|       |          |             |

1. Further information concerning other than nominal feed rates, feed rates for running-in and for part load operation, and recommended oil types, is given in instruction book Volume I, OPERATION.

2. The cylinder lubricators are mounted on the front end of the engine.

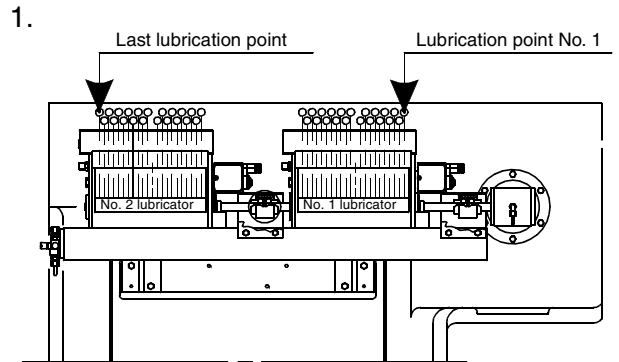
For lubricator number. See the sketch.

3. Each cylinder has six injection points. Check that the pumps for all injection points are activated during one revolution of the engine.

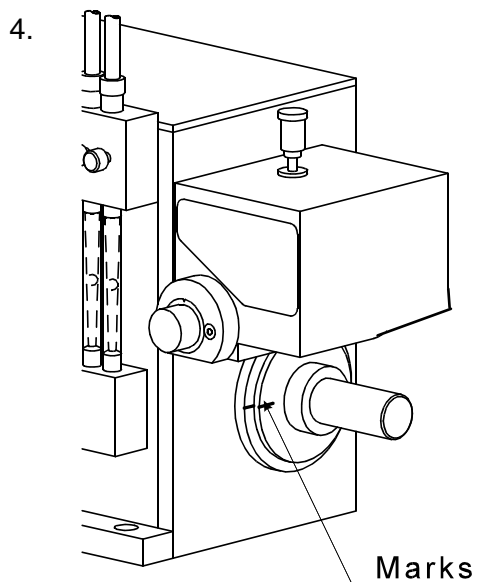
4. Turn the engine in AHEAD direction until the mark on No. 1 lubricator housing is in line with the mark on the scale at the lead-in shaft. This indicates that the pump stroke is completed for lubricating point No. 1.

5. Check that the crankthrow for cylinder No. 1 is positioned between **D03-22** and **D03-23** degrees past BDC.

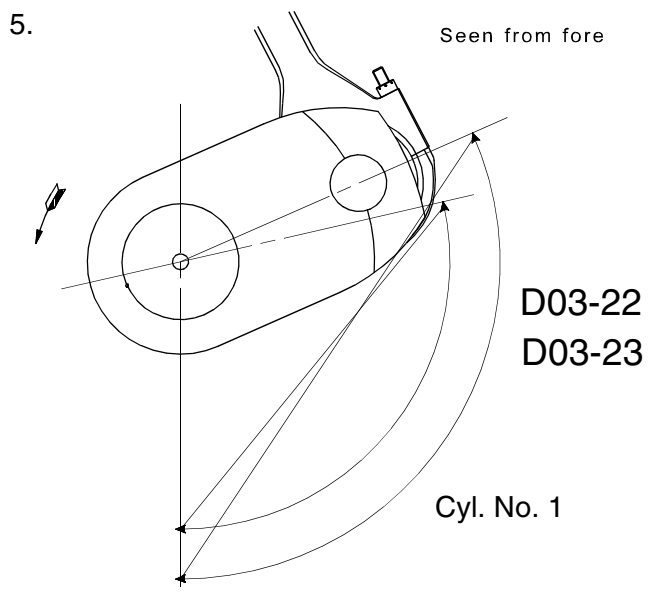
Use the numbers on the turning wheel to determine the exact position of the crankthrows.



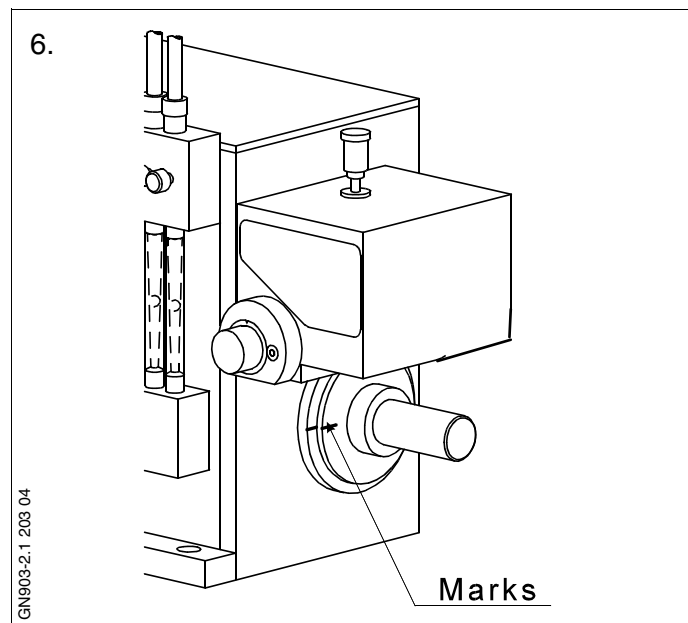
GN903-2.1 227 01



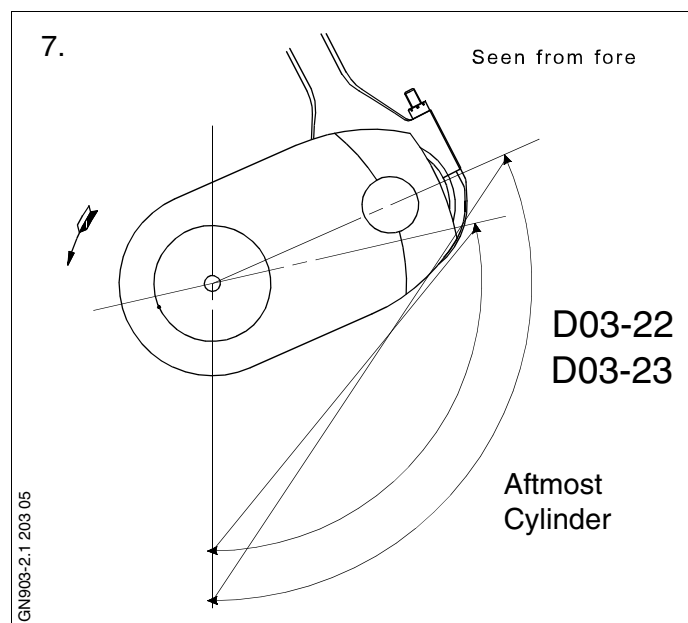
GN903-2.1 203 04



GN903-2.1 203 05



6. Turn the engine further in AHEAD direction until the mark on the second lubricator housing is in line with the mark on the scale at the lead-in shaft. This indicates that the pump stroke is completed for the last lubricating point on No. 2 lubricator.
7. Check that the crankthrow for the aftmost cylinder is positioned between **D03-22** and **D03-23** degrees past BDC.
8. If one of the crankthrows is not in the correct position, adjust the lubricators.  
*See Procedure 903-2.3.*



## Note!

Any adjustment of lubricator No. 1 will alter the setting of lubricator No. 2.

Therefore, any adjustment of lubricator No. 1 **must** be followed by a re-adjustment of lubricator No. 2.

1. Remove the guards over the couplings.

Loosen the set screws in the couplings and pull the coupling wheels out of contact with the planet wheels.

2. For adjustment of lubricator No. 1, turn the crankshaft in AHEAD direction until the crankthrow for cylinder No. 1 is **D03-24** degrees past BDC for lubricator No. 1.

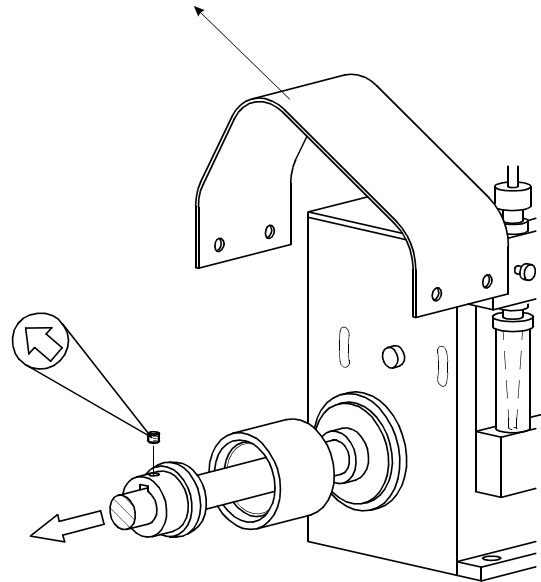
If necessary, turn the lubricator shaft in AHEAD direction.

Push the coupling wheels in to contact with the planet wheel for lubricator No. 1 and fasten the screws.

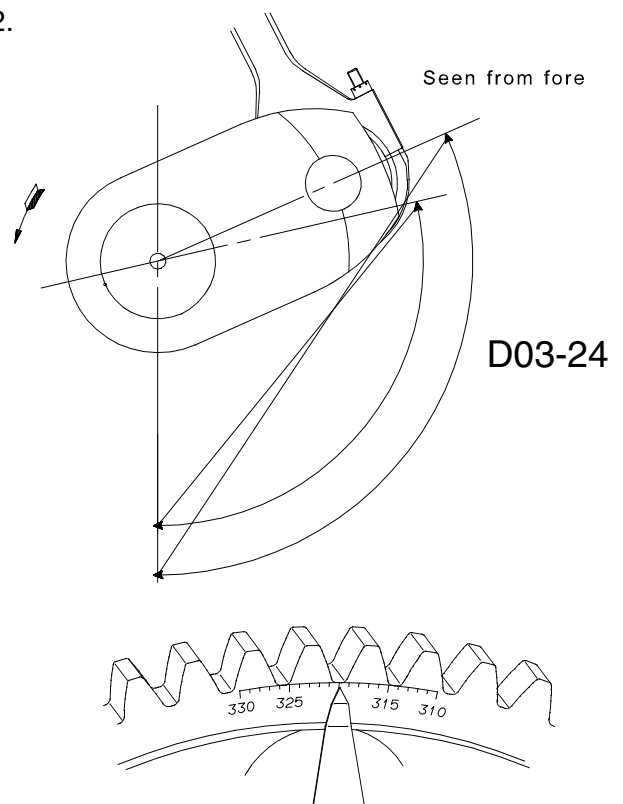
For adjustment of lubricator No. 2, turn the crankshaft in AHEAD direction until the crankthrow for the aftmost cylinder is **D03-24** degrees past BDC.

Use the numbers on the turning wheel to determine the exact position of the crankthrows.

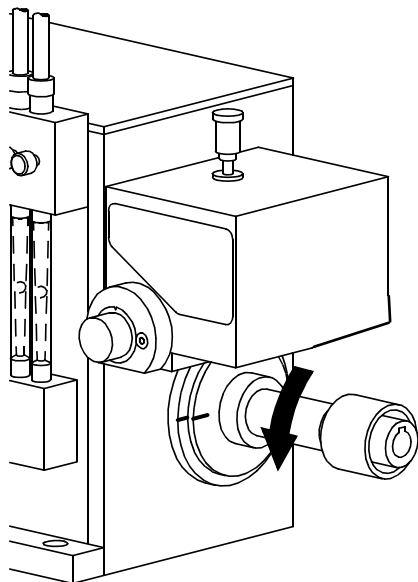
1.



2.



3.

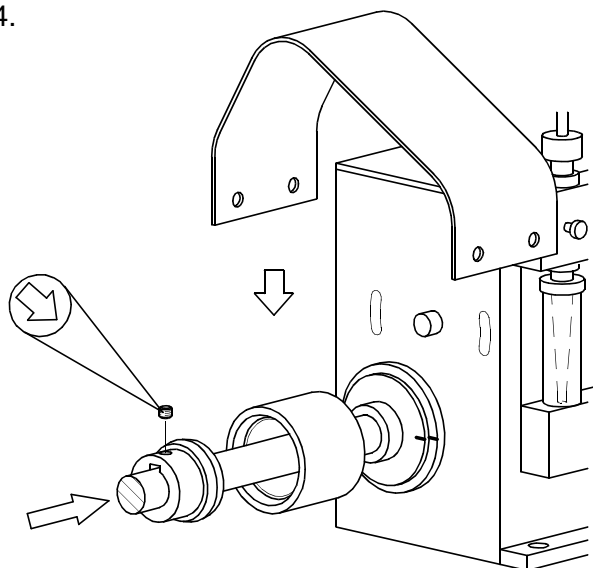


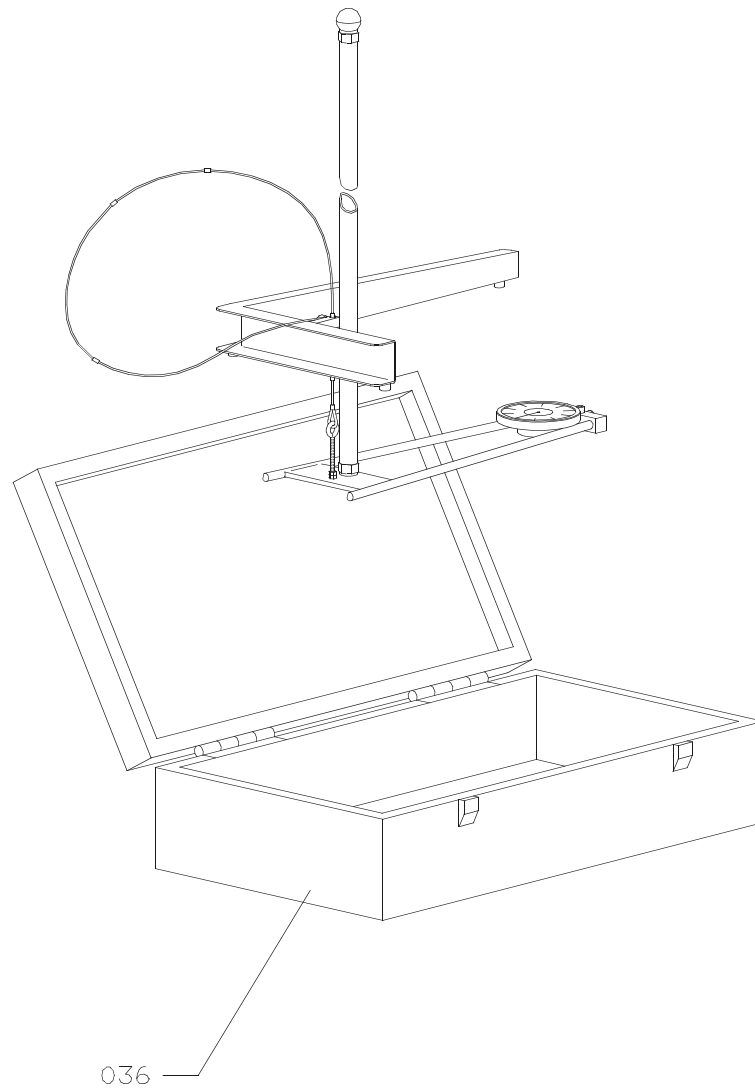
3. Turn the pump shaft slowly in AHEAD direction until the mark on the scale at the lead-in shaft is in line with the mark on the lubricator housing.
4. Re-engage the coupling wheel and tighten the set screw.

Re-check the setting.  
*See Procedure 903-2.1.*

Mount the guard over the coupling.

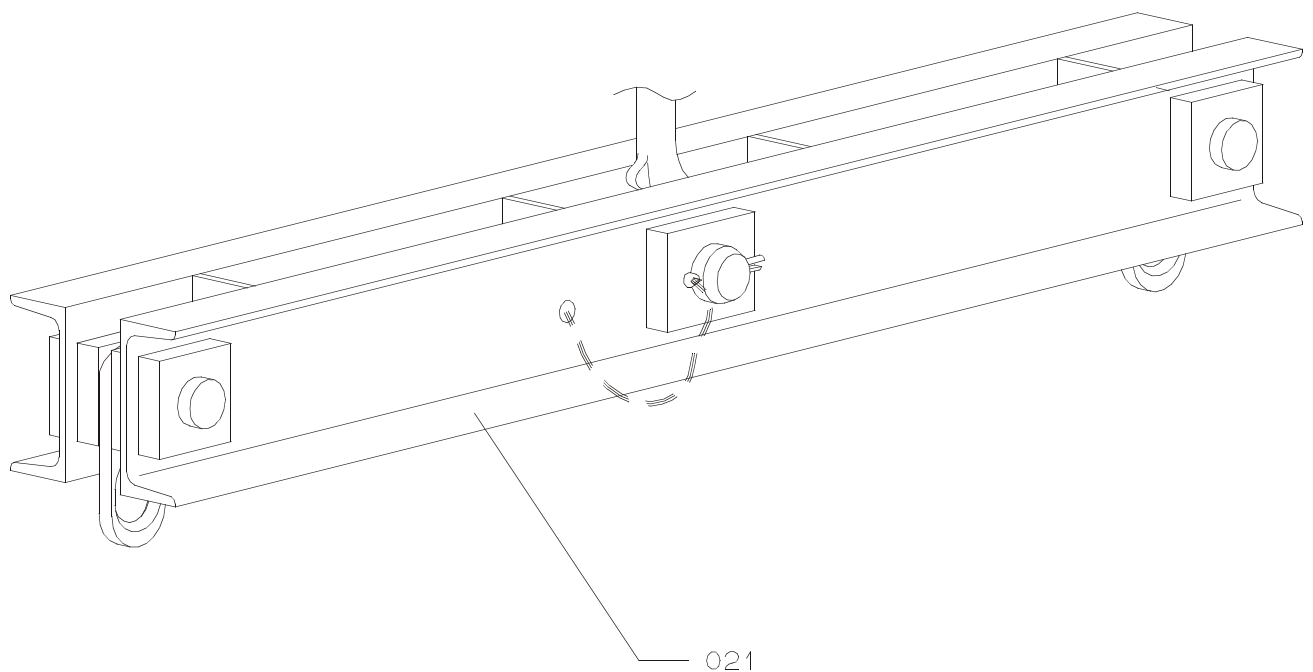
4.





| Item No. | Item Description         |
|----------|--------------------------|
| 036      | Cylinder gauge, complete |

| Item No. | Item Description |
|----------|------------------|
|          |                  |



| Item No. | Item Description |
|----------|------------------|
|          |                  |